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Galactic Chemical Evolution
Model of the Milky Way

Autor:
Alejandro Huerga Rodriguez

Tutores: Yago Ascasibar Sequeiros y Pablo Corcho Caballero

Universidad Autónoma de Madrid
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Abstract

Galaxy evolution is fundamentally driven by the interplay between star formation, supernova feedback, radiative cooling, and gas dynamics. While star formation histories require a continuous supply of gas from the circumgalactic medium, classical closed-box models fail to explain the observed deficit of metal-poor stars and the persistence of star formation over gigayear timescales. In this work, we implement a numerical chemical evolution model for a Milky Way-like galaxy based on concentric cylindrical shells in hydrostatic equilibrium. The model incorporates a self-regulating energy balance—assuming equipartition between cosmic rays, magnetic fields, and turbulence—and incorporates energy dissipation through a radiative cooling timescale, t_c , derived from the analytical framework of the Sedov-Taylor phase and an exponential gas infall ($I = I_0 \exp(-t/\tau)$).

The simulated radial profiles of stellar surface density, gas surface density, star formation rate, and gas depletion time show good agreement with the observational data from Mollá et al. [1] across most of the disk. A fixed infall timescale already reproduces the profiles well at outer radii ($r \gtrsim 5$ kpc); the adoption of an inside-out formation scenario with a radially dependent timescale, $\tau(r) = 1000 + 1500 (r/\text{kpc})$ Myr, extends this agreement down to $r \sim 1$ kpc, significantly improving the match to the observed star formation rate profile across the inner disk. The model successfully maintains a Toomre stability parameter $Q \geq 1$ throughout most of the galactic history, demonstrating how supernova feedback prevents runaway gravitational collapse while the gradual depletion of the gas reservoir leads to a long-term decline in star formation. However, the model yields disk scale heights z_0 significantly below the observed range of 100–1000 pc, and systematically overestimates the stellar density at small radii ($r \lesssim 5$ kpc). These discrepancies are attributed to the assumption of a constant star formation efficiency, the plane-parallel potential geometry, and the exclusion of inner-disk processes such as bar-driven radial gas inflows and AGN feedback. Overall, this model underscores the critical role of supernova feedback and inside-out formation in the self-regulation and structural evolution of the Milky Way.

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1 Introduction

The Milky Way is a unique galaxy in the sense that it is the only one in which individual stellar populations and gas clouds can be resolved at parsec precision. For external galaxies such as M31 or more distant systems, only integrated values over the whole galaxy or averages over wide spatial zones are available. On the other hand, thanks to catalogues such as *Gaia* [2] and the radio cartography used by Mollá et al. [1], radial profiles can be constructed with a precision unmatched by any other galaxy. Even though the Milky Way is not the only system in which the stellar surface density (Σ_*) and gas surface density (Σ_{gas}) can be measured, it is the only one where these quantities are independent and direct. For external galaxies, Σ_* is estimated from the mass-luminosity relation, which introduces larger uncertainties. In the Milky Way, by contrast, individual stars can be counted and the local dynamics can be measured with high precision.

The Milky Way is classified, following the Hubble sequence [3], as a barred spiral galaxy (SBbc), where S indicates a spiral morphology (flattened disk, spiral arms, and young stellar populations), B indicates the presence of a central bar, and bc denotes an intermediate degree of arm winding. Furthermore, since physical properties depend primarily on radius and height, the system naturally lends itself to a cylindrical shell approximation under the assumption of axial symmetry.

1.1 Chemical evolution and star formation

In order to understand the chemical evolution of galaxies, the closed-box model was proposed by Searle and Sargent [4] as the simplest approximation. This model consists of a single isolated ring with no external supply of mass (infall) and no mass expulsion (outflows), composed initially of hydrogen and helium. Metals ejected by stars are assumed to mix instantaneously with the surrounding gas, and instantaneous recycling of stellar material is also assumed. This model yields a metallicity (Z) given by:

$$Z = y \cdot \ln(1/f_g) \quad (1.1)$$

where y is the stellar yield (the mass of metals released per generation of stars) and f_g is the gas mass fraction. However, the closed-box model predicts a higher number of metal-poor stars than what is actually observed — a discrepancy known as the G-dwarf problem [5]. This excess arises because, in a closed system, a large fraction of the initial gas must be consumed before the metallicity can rise significantly, resulting in a high number of stars born with low metallicity. Observations show, however, that such stars are much scarcer than predicted. This naturally motivates the inclusion of gas infall: a continuous supply of metal-poor gas delays the rapid rise of metallicity in the gas phase while keeping the star formation rate low at early

times, producing the observed deficit of metal-poor stars at early epochs and the gradual enrichment seen today. The mass evolution is therefore governed by the following system of ordinary differential equations (ODEs):

$$\begin{cases} \dot{M}_g = I - \psi \\ \dot{M}_* = \psi \end{cases} \quad (1.2)$$

where M_g is the gas mass, M_* the stellar mass, I the infall rate, and ψ the star formation rate — the amount of stellar mass formed per unit time. Based on the observational relation established by Schmidt [6] and extended by Kennicutt [7], $\Sigma_{\text{SFR}} \propto \Sigma_{\text{gas}}^n$ with $n \approx 1.4$, the following prescription for ψ is adopted in this work:

$$\text{SFR} \equiv \psi = \frac{M_{\text{fe}}}{t_{\text{ff}}} \quad (1.3)$$

where M_{fe} is the fraction of gas mass that collapses into stars and t_{ff} is the free-fall time — the characteristic timescale for a gas cloud to collapse under its own gravity. The fraction M_{fe} is determined by the star formation efficiency (SFE), defined as the fraction of the gas mass converted into stars per free-fall time:

$$\text{SFE} \equiv \frac{M_{\text{fe}}}{M_g} \quad \Rightarrow \quad M_{\text{fe}} = \text{SFE} \cdot M_g \quad (1.4)$$

so that $\psi = \text{SFE} \cdot M_g / t_{\text{ff}}$. Observationally, SFE is found to be of order 1–2% per free-fall time in molecular clouds [7, 8], reflecting the fact that only the densest subregions of a molecular cloud — those that exceed the local collapse threshold — actually form stars, while the bulk of the gas is supported by turbulence, magnetic fields, and thermal pressure.

The free-fall time is derived from the equation of motion:

$$\frac{d^2 \mathbf{r}}{dt^2} = -\frac{1}{\rho} \nabla P - \nabla \Phi \quad (1.5)$$

where $\mathbf{r}$ is the position vector, P the pressure, and Φ the gravitational potential. By neglecting the pressure gradient ($\nabla P \approx 0$) — appropriate for a pressureless cold collapse of a spherical cloud — one obtains:

$$t_{\text{ff}} = \left(\frac{3\pi}{32G\rho_0} \right)^{\frac{1}{2}} \quad (1.6)$$

where G is the gravitational constant and ρ_0 is the central gas density.

A more physically motivated approach to the SFE would account for magnetic fields and turbulence through a probability density function (PDF) of density fluc-

tuations. Since molecular clouds are characterised by efficient radiative cooling and highly supersonic turbulent flows, they can be modelled as isothermal media, in which density fluctuations follow a log-normal distribution arising from the stochastic nature of supersonic shocks [9, 10]. In this framework, the relevant collapse threshold is the Bonnor–Ebert mass [11]:

$$M_{\text{BE}}(\rho) = 1.182 \frac{\left(c_s^2 + \frac{3}{2}v_A^2\right)^{3/2}}{\sqrt{G^3\rho}} \quad (1.7)$$

where c_s is the sound speed, ρ the cloud density, and v_A the Alfvén velocity defined as $v_A^2 = B^2/(4\pi\rho)$ with B the magnetic field strength. Nevertheless, in this work a constant SFE of order 0.4–2% is adopted for simplicity, as this approximation has been widely used in semi-analytic chemical evolution models and provides a good first-order description of the global star formation history [12, 1]. The implications of this choice are discussed in Sections 2 and 4.

1.2 Supernova feedback and disk stability

Once the mechanism driving star formation is understood, it is necessary to consider the energy balance. For stars with $M_* \gtrsim 8 M_\odot$, once the nuclear fuel in the core is exhausted, radiation pressure can no longer counterbalance gravity. The iron core collapses on a dynamical timescale of ~ 1 s, rebounding as a shock wave that ejects the stellar envelope in a core-collapse supernova, releasing approximately $E_0 \sim 10^{51}$ erg, of which only $\sim 1\%$ couples to the surrounding ISM as mechanical energy [13]. This feedback is fundamental: an increase in energy raises the velocity dispersion (σ), which increases the disk scale height (z_0), as will be demonstrated in Section 2. Without a compensating mechanism, this would lead to an indefinitely expanding disk — inconsistent with the thin-disk scale heights of 100–1000 pc observed in nearby spirals [14]. This implies the existence of an opposing process that prevents unbounded expansion.

This regulatory component is the radiative cooling that occurs during the late stages of supernova remnant evolution. When a supernova occurs, the ejecta undergo a free expansion phase followed by an adiabatic expansion phase. The latter begins when the mass of swept-up ISM becomes comparable to the initial ejecta mass, and is known as the Sedov–Taylor phase [15, 16]. It is characterised by the deceleration of the shock wave as the gas is heated, with negligible radiative losses compared to the mechanical energy transferred to the ISM. After this phase, the gas cools rapidly and begins to radiate efficiently. The timescale at which this radiative transition occurs, t_c , was derived by Calleja [17] and is adopted in this work, as it provides the characteristic cooling timescale needed to close the energy balance. Accounting for both energy injection and dissipation, the total energy of the gas is defined as

the sum of its turbulent kinetic and potential energy plus the supernova feedback, minus the energy lost to stellar processes and radiation.

The structural evolution of a galactic disk is therefore governed by the competition between gravity, which drives collapse, and stabilising mechanisms such as differential rotation and internal pressure (thermal and turbulent). A standard metric for assessing the local gravitational stability of a gaseous disk is the Toomre Q parameter [18]:

$$Q_g = \frac{\sigma_v \kappa}{\pi G \Sigma_g} \quad (1.8)$$

where Σ_g is the gas surface density, G the gravitational constant, and σ_v the effective velocity dispersion (accounting for both thermal and turbulent contributions). The epicyclic frequency κ quantifies the stabilising effect of differential rotation:

$$\kappa^2 = \frac{2\Omega}{R} \frac{d}{dR} (R^2 \Omega) = 4\Omega^2 \left(1 + \frac{R}{2\Omega} \frac{d\Omega}{dR} \right) \quad (1.9)$$

where Ω is the angular velocity and R is the radii. The stability of the disk is determined by the value of Q :

- If $Q < 1$, the disk is unstable to large-scale radial perturbations, leading to the formation of clumps, molecular clouds, and prominent spiral structures.
- If $Q \geq 1$, the disk is stable: rotation on large scales and velocity dispersion on small scales jointly counteract gravity.

Observations suggest that galaxies self-regulate to maintain $Q \approx 1$: when star formation is intense, supernova feedback increases σ_v , raising Q and stabilising the disk [8]. Conversely, if the disk becomes too stable, star formation drops, turbulence dissipates, and Q decreases until instabilities grow again. This feedback loop explains why observed star formation rates are much lower than simple free-fall estimates would predict [7, 6], and motivates the coupled dynamical model developed in this work.

1.3 Motivation and objectives of this work

Given these mechanisms, chemical evolution models such as those of Chiappini, Matteucci, and Gratton [12] and Mollá et al. [1] have demonstrated that an exponential infall $I = I_0 e^{-t/\tau}$ with $\tau = \tau(r)$ provides a better fit to the observed radial profiles. This inside-out growth scenario is fundamental to explaining the earlier assembly of the inner disk relative to the outer disk [19]. However, these models do not account for vertical disk dynamics. Models that do incorporate vertical dynamics, such as Ostriker, McKee, and Leroy [8], have not always been compared systematically with radial profiles of the Milky Way. This gap motivates the present work.

In this work, a numerical model of concentric cylindrical shells in hydrostatic equilibrium is implemented, with exponential gas infall and supernova feedback regulated by the cooling timescale t_c from Calleja [17]. The stellar surface density (Σ_*), gas surface density (Σ_{gas}), star formation rate surface density (Σ_{SFR}), and gas depletion time (τ_{dep}), which indicate how long the galaxy would take to consume its gas reservoir at the current star formation rate ($\Sigma_{\text{gas}}/\Sigma_{\text{SFR}}$), are systematically compared with the observational data of Mollá et al. [1]. The main goal is to determine how well a dynamically coupled but physically simple model can reproduce the observed radial profiles of the Milky Way, and to identify which additional ingredients are required.

2 Methods

2.1 Theoretical foundations

Physical model and vertical structure

In spiral galaxies like the Milky Way, most of the visible mass is concentrated in a thin disk, such that the following hold:

- **Axial symmetry:** Due to rotation, physical properties such as gas density and metallicity vary much more significantly with radius (r) and height (z) than with the azimuthal angle (ϕ). Thus, dividing the galaxy into independent concentric rings is a standard approximation.
- **Time scales:** The radial mixing timescale is much longer than the processing time within each ring; therefore, except for gas infall, each ring can be considered a closed system.

Since the disk thickness is much smaller than its radius ($h \ll R$), vertical forces are dominated by the local mass density—that is, the gas and stars in the solar neighbourhood. At small scales, the gravitational field behaves similarly to an infinite plane:

$$\Phi = g|z| \tag{2.1}$$

and, consequently, the acceleration is given by:

$$g(z) = \begin{cases} +g & \text{if } z > 0 \\ 0 & \text{if } z = 0 \\ -g & \text{if } z < 0 \end{cases} \tag{2.2}$$

Thus, the surface potential energy density can be determined as:

$$u = \int_{-\infty}^{\infty} \rho \Phi dz \quad (2.3)$$

To evaluate this, an expression for ρ is required. This relationship can be derived by considering that observations indicate galaxies maintain their vertical structure over long periods—meaning they are systems in equilibrium. Therefore, the force balance between pressure and gravity allows us to use the hydrostatic equilibrium condition:

$$\frac{1}{\rho} \frac{\partial P}{\partial z} = -g \quad (2.4)$$

where the pressure is assumed to be isotropic and equal to $\rho\sigma^2$. This formulation accounts for the total effective pressure, including not only the turbulent motion of the gas but also the non-thermal support provided by magnetic fields and cosmic rays, which are typically of the same order of magnitude in the Galactic disk [20]. Solving this equation yields:

$$\rho = \rho_0 e^{\frac{-z}{z_0}} \quad ; \quad z_0 = \frac{\sigma^2}{g} \quad (2.5)$$

Furthermore, by normalisation, the total gas mass (M_g) must satisfy:

$$M_g = \int_{-\infty}^{\infty} \rho dV \Rightarrow \rho_0 = \frac{M_g}{2Az_0} = \frac{\Sigma_g}{2z_0} \quad (2.6)$$

Applying Gauss's Law:

$$\oint g dA = -4\pi GM \Rightarrow g = 2\pi\Sigma G \quad (2.7)$$

the total potential energy U can be determined as:

$$U = uA = M_g g z_0 \quad (2.8)$$

If we also consider energy equipartition ($E_{\text{turb}} \approx E_{\text{CR}} \approx E_{\text{mag}}$), then $E_k = \frac{3}{2}M_g\sigma^2$, so the total energy is:

$$E = E_k + U = \frac{5}{2}M_g\sigma^2 \quad (2.9)$$

leading to:

$$\sigma^2 = \frac{2}{5} \frac{E}{M_g} \quad ; \quad v_{\text{esc}}^2 = \frac{4}{5} \frac{E}{M_g} \quad (2.10)$$

where v_{esc} is a characteristic vertical velocity obtained by equating the kinetic energy

to the gravitational potential difference across one scale height,

$$\frac{1}{2}mv_{\text{esc}}^2 \sim m\Delta\Phi \quad \Rightarrow \quad v_{\text{esc}} \approx \sqrt{2gz_0}. \quad (2.11)$$

Radiative cooling and the Sedov-Taylor framework

The energy radiated by the gas must also be considered to maintain a realistic energy balance. To this end, we adopt a cooling timescale t_c derived from the analytical framework of the Sedov-Taylor phase [15, 17]. In this phase, the expansion of the supernova remnant is assumed to be adiabatic, implying that energy is conserved and follows the relation $R^3v_s^2 \sim \text{const}$. By utilizing this theoretical background, we can define the transition to the radiative phase, where gas cooling becomes dominant and energy is rapidly dissipated. The Sedov-Taylor solution gives:

$$v_s = \frac{2}{5} \left(\frac{\chi E_0}{\rho_0} \right)^{\frac{1}{5}} t^{3/5} \quad (2.12)$$

By considering the shock wave pressure as $P_1 = \frac{3}{4}\rho_0v_s^2$ and the post-shock number density as $n_1 = 4n_0$, we can determine t_Λ , the characteristic cooling time of the remnant:

$$t_\Lambda = \frac{u_1}{|\dot{u}_{c,1}|} = \frac{3P_1}{2n_1^2\Lambda(T)} \quad (2.13)$$

where $\Lambda(T)$ is the radiative cooling function, approximated in the temperature range $T \in [10^5, 10^7]$ K as a power law [13]:

$$\Lambda(T) \approx \Lambda_a \left(\frac{T}{T_a} \right)^a \quad (2.14)$$

with normalisation $\Lambda_a = 10^{22} \text{ erg cm}^3 \text{ s}^{-1}$, reference temperature $T_a = 10^5$ K, canonical supernova energy $E_0 = 10^{51}$ erg, and slope $a = -0.9$.

To obtain the cooling time t_c , we must find the time at which the supernova remnant (SNR) becomes radiative by solving:

$$\left. \frac{\partial(t + t_\Lambda)}{\partial t} \right|_{t_c} = 0 \quad (2.15)$$

Solving for t_c yields the following expression, based on Calleja [17, Eq. 2.20]:

$$t_c^{\frac{1}{5}(11-6a)} = \frac{81(1-a)m_H}{1600n_0\Lambda_a} \left(\frac{9m_H}{80k_B T_a} \right)^{-a} \left(\frac{16\chi E_0}{375n_0m_H} \right)^{\frac{2}{5}(1-a)} \quad (2.16)$$

where the parameters are defined as follows:

- $m_H = 1.67 \times 10^{-24}$ g: hydrogen mass, defining the characteristic mass scale of

the medium¹.

- $n_0 = 1 \text{ cm}^{-3}$: ambient number density of the ISM into which the supernova remnant expands.
- $k_B = 1.38 \times 10^{-16} \text{ erg K}^{-1}$: Boltzmann constant.
- $T_a = 10^5 \text{ K}$: reference temperature at which the cooling function is normalised.
- $\Lambda_a = 10^{-22} \text{ erg cm}^3 \text{ s}^{-1}$: normalisation of the cooling function at T_a .
- $\chi \approx 2$: dimensionless geometric factor arising from the exact Sedov–Taylor solution [15].
- $a = -0.9$: power-law slope of the cooling function in the relevant temperature range.

This formulation provides the characteristic timescale for radiative dissipation, effectively completing the necessary components to describe the energy balance of the system.

System of differential equations

With these quantities defined, the system of differential equations for the masses is:

$$\begin{cases} \dot{M}_g = I - \psi \\ \dot{M}_* = \psi \end{cases}$$

The ODE for the energy evolution is expressed as:

$$\dot{E}_g = I \left(\frac{2}{5} \frac{E}{M_{\text{gas}}} + \frac{1}{2} v_0^2 \right) + \dot{E}_{\text{SN}} - \psi \frac{E}{M_g} - L \quad (2.17)$$

where I is the gas infall rate [$M_\odot \text{ Myr}^{-1}$], $\dot{E}_{\text{SN}}$ is the energy injected by supernovae per unit time, $L = E/t_{\text{cooling}}$ represents radiation losses, and ψ is the Star Formation Rate (SFR), modelled as:

$$\psi = \nu M_g \quad (2.18)$$

Here, $\nu = \text{SFE}/t_{ff}$ [Myr^{-1}] is the star formation efficiency per free-fall time, and t_{ff} is derived by neglecting pressure in the hydrostatic equilibrium equation:

$$t_{ff} = \left(\frac{3\pi}{32G\rho_0} \right)^{\frac{1}{2}}$$

¹An additional factor m_H has been included in the denominator of the last term in Equation (2.16). This correction addresses a transcription error identified in the original source, ensuring that the cooling timescale remains dimensionally consistent with the mass density ρ utilized in our numerical implementation.

Physically, the collapse threshold is set by the Jeans mass, above which thermal pressure cannot support the cloud against gravity. However, a full computation of the collapsing mass fraction is beyond the scope of this work. Instead, a constant efficiency SFE is adopted. The Toomre parameter (Q), which describes the gravitational stability of the disk [18], is defined as:

$$Q = \frac{c_s \kappa}{\pi G \Sigma}$$

where c_s is the sound speed and κ is the epicyclic frequency:

$$\kappa^2 = \frac{2\Omega}{R} \frac{d}{dR} (R^2 \Omega)$$

Star formation efficiency and supernova energy injection

In this model, the SFE was chosen to be constant, despite the discussion about its dependence on the log-normal density function, which depends on the velocity dispersion. In other words, a more precise simulation would need to take into account its dependence, which would imply coupling the velocity dispersion to the ODE system, but this is beyond the scope of this work. Nevertheless, the constant SFE approximation has been constantly used in semi-analytic chemical evolution models and provides a good first-order description of the global star formation history [12, 1].

The supernova injection of energy is given by:

$$\dot{E}_{\text{SN}} = \eta_{\text{SN}} \psi E_0 \quad (2.19)$$

where η_{SN} [$\text{M}_{\odot}^{-1}$] is the number of core-collapse supernovae per unit stellar mass formed, and $E_0 = 10^{51}$ erg is the canonical energy released per supernova. The parameter η_{SN} is deeply connected with the Initial mass function (IMF) since η_{SN} determines the number of supernovae per unit of stellar mass, which is given by the number of stars that become unstable against gravitational collapse at the end of their lives. In this study were considered stars with $M_* \geq 8 \text{ M}_{\odot}$, which, alongside observations, corresponds to, approximately, 1–2% of the total stellar mass formed, giving a $\eta_{\text{SN}} \approx 0.01 \text{ M}_{\odot}^{-1}$. This value is consistent with observational estimates of the core-collapse supernova rate in nearby spiral galaxies [13]. Therefore, the supernovae feedback is a source of kinetic energy for the gas in galaxies, helping to increase the velocity dispersion and, thus, the vertical support as follow: $z_0 \propto \sigma^2$, as well as the reduction of ρ_0 once $\rho_0 \propto z_0^{-1}$. That mechanism illustrates clearly how the supernovae feedback contributes to the stability and self regulation of the galaxy.

2.2 Numerical simulation

Numerical implementation and algorithm

The model has been implemented in Python, with the code structured around a main class that encapsulates the physical parameters and solver methods [21, 22].

Given the coupled nature of the evolution equations, two different adaptive-step solvers were employed to ensure numerical stability and convergence across different physical regimes. For the radial profiles with constant accretion timescales, a Runge-Kutta method (RK45) was selected. This explicit solver offers a robust balance between computational cost and accuracy, providing a local truncation error of order $\mathcal{O}(h^5)$ and proving particularly stable during the rapid initial gas infall phase. However, for more complex scenarios, such as the implementation of variable accretion timescales $\tau(r)$ and high-resolution temporal studies, the LSODA (Livermore Solver for Ordinary Differential Equations) was adopted. This solver monitors the stiffness of the equations; when the system is non-stiff, it uses Adams–Moulton methods (similar to Runge-Kutta but more efficient), automatically adjusting the time step dt .

The process follows an iterative scheme:

1. **Initialisation:** Initial conditions are defined at $t = 0$ for $M_{g,0}$, $M_{*,0}$ and σ_0 .
2. **Slope calculation:** At each time step dt , the rate of change is evaluated at four intermediate points.
3. **Variable update:** M_g , M_* , and E_g are updated, and the evolution of the masses and σ is computed.
4. **Diagnostic calculations:** Following the mass and energy update, the Toomre parameter Q , the scale height z_0 , and the relative disk thickness z_0/R are computed.
5. **Radial grid calculation:** A vector of radii $r \in [r_i, r_f]$ kpc is generated with a spacing of 1 kpc.

Initially, a negligible mass of $M_{g,0} = 1 \times 10^4 M_\odot$ and $M_{*,0} = 1 \times 10^{-6} M_\odot$ was introduced to avoid numerical errors, alongside a constant infall of $1 \times 10^6 M_\odot \text{ Myr}^{-1}$ and a initial velocity dispersion $\sigma_0 = 15 \text{ km/s}$, to verify the validity of the integration method.

Next, an exponential infall was implemented as $I = I_0 e^{-t/\tau}$, so that the radial surface density profile takes the form $\Sigma = \Sigma_0 e^{-r/R_d}$. The adopted parameters were $\tau = 12\,000 \text{ Myr}$, $R_d = 3000 \text{ pc}$, and $\Sigma_0 = 2000 M_\odot \text{ pc}^{-2}$. The normalization constant I_0 is determined by requiring that the total mass accreted from the beginning of the

galaxy's formation up to the current time (t_G) matches the observed total mass M_{total} :

$$\int_0^{t_G} I_0 e^{-t/\tau} dt = I_0 \tau (1 - e^{-t_G/\tau}) = M_{\text{total}} \quad (2.20)$$

where t_G represents the current age of the Galactic disk. This ensures that the model is calibrated to current observations. The integration was subsequently applied across the radial range $r \in [0.001, 22]$ kpc where $r = 0.001$ kpc serves as a numerical proxy for the galactic center in order to avoid numeric errors, and the resulting Σ_* , Σ_{gas} , Σ_{SFR} , and τ_{dep} were compared with the data from Mollá et al. [1].

Following this, the model was further refined to provide a more realistic description of the formation of the Milky Way. An initial gas mass of $M_{g,0} = 1 \times 10^6 M_\odot$ —negligible yet non-zero to avoid numerical errors—was introduced together with $M_{*,0} = 0 M_\odot$ and $\sigma_0 = 15$ km/s. The inside-out formation scenario was activated [12, 19], with τ increasing linearly with radius as:

$$\tau(r) = 1000 + 1500 (r/\text{kpc}) \quad [\text{Myr}] \quad (2.21)$$

for $r \in [0.001, 22]$ kpc, so that the inner disk assembles on a shorter timescale than the outer disk.

Finally, calculations were performed for two extreme values, $\tau = 1000$ Myr and $\tau = 1 \times 10^9$ Myr, to study how the system behaves when mass injection is almost instantaneous versus quasi-constant over a Hubble time.

3 Results and Discussion

3.1 Verification calculations

Firstly, z_0 and ρ_0 were plotted to analyse their behaviour. The results are shown in Figure 1, where it can be observed that the scale height increases with σ . Thus, the more turbulent the medium, the thicker the disk, which is consistent with observations of edge-on spiral galaxies [14, 23]. It is also notable that the central gas density decreases rapidly as σ increases.

3.2 Constant Infall Model

Next, the cooling time derived from Sedov-Taylor was implemented [15, 16, 24], together with the galactic model for a constant infall and a negligible initial mass ($M_{g,0} = 1 \times 10^4 M_\odot$ and $M_* = 1 \times 10^{-6} M_\odot$). The results are presented in Figure 2, where the gas mass (M_g) barely changes after 2000 Myr, reaching a quasi-stationary state. Furthermore, around 3000 Myr, the star mass (M_*) overcomes the gas mass (M_g) and continues to grow indefinitely since the infall never stops in this case.

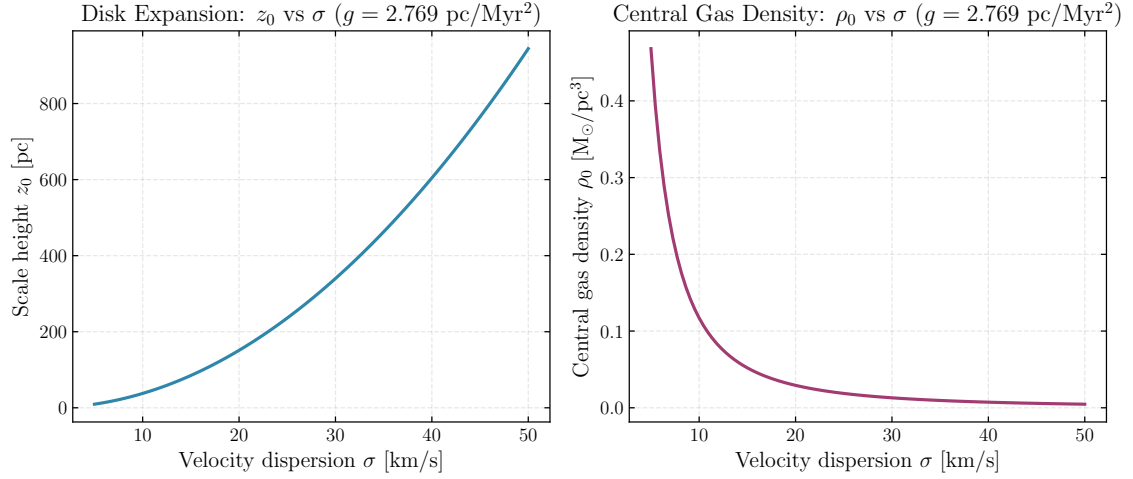


Figure 1: Scale height (z_0) and central gas density (ρ_0) as a function of velocity dispersion σ .

Additionally, as evidenced in Figure 2, the velocity dispersion (σ) has a pronounced drop in the early stages and then reverses its trend, starting to increase very slowly until reaching an equilibrium around 5 km/s. This behaviour is consistent with results from similar models [10].

To verify that this behaviour satisfies the stability condition ($Q \geq 1$) and a realistic disk thickness (z_0), both quantities are plotted in Figure 3. The model remains gravitationally stable throughout the age of the Universe, with $Q > 1$ at all times. However, it is worth noting that Q begins to rise steadily after ~ 2000 Myr, reflecting the gradual depletion of the gas reservoir: as M_g decreases, the surface density Σ drops, and from Equation (1.8) it follows that Q increases. If this trend continued indefinitely, the galaxy would eventually become entirely inert—unable to form new stars—although the constant infall prevents a full shutdown in this model.

Regarding the disk thickness, the model yields an extremely thin disk: z_0 settles well below the typical scale height of 100–1000 pc observed in nearby edge-on spirals [14]. This can be understood from Equations (2.5) and (2.7). The scale height is $z_0 = \sigma^2/g$, where $g = 2\pi G\Sigma \propto M_{\text{total}}/A$. As shown in Figure 2, σ reaches a quasi-stationary value after the initial transient, while M_{total} continues to grow due to the uninterrupted infall. The resulting increase in g drives z_0 downward over time. This indicates that a purely constant infall—without a temporal decay (τ) to limit the total accreted mass—would lead to a disk that continues to thin and contract indefinitely. Such a runaway process would fail to reproduce the stable scale height and mass distribution observed in Milky Way-like galaxies.

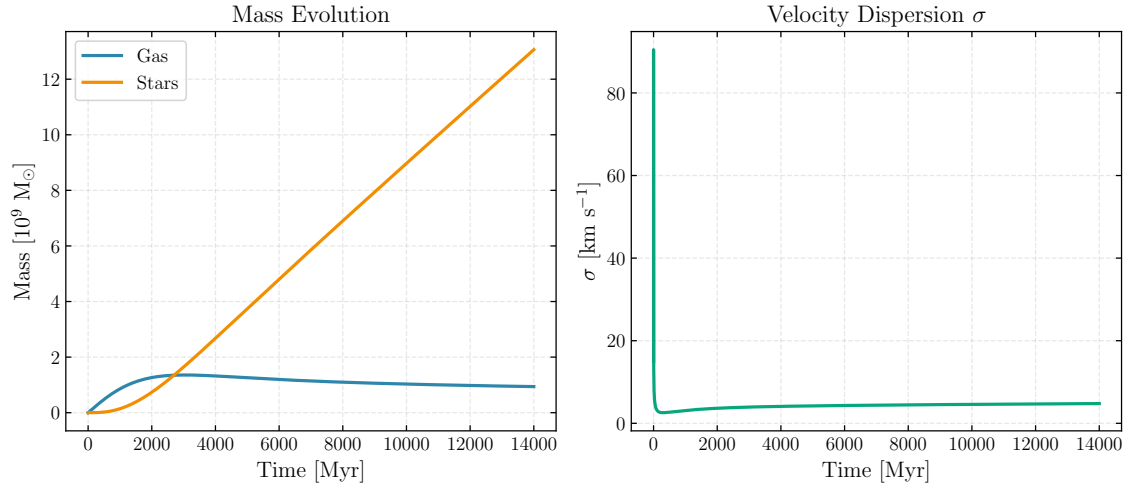


Figure 2: Mass evolution of gas and stars (left), and time evolution of the velocity dispersion σ (right) for the constant infall model.

3.3 Exponential infall model

Moving to a more realistic simulation of the Milky Way, a negligible initial gas and stellar mass was adopted once more, together with an initial velocity dispersion of 15 km s^{-1} and an exponential infall with $\tau = 12\,000 \text{ Myr}$. A low star formation efficiency ($\text{SFE} = 0.004$) and $\eta_{\text{SN}} = 0.01$ were also assumed, along with a central surface density $\Sigma_0 = 2000 \text{ M}_\odot \text{ pc}^{-2}$ and a disk scale radius $R_d = 3000 \text{ pc}$. This value of Σ_0 is representative of the high-density environments found in the inner regions of a Milky-Way-like disk [1, 12]. It was chosen to ensure that the model reaches a final state consistent with current observational constraints, allowing for a direct comparison between the simulated evolution and the present-day Galaxy. In this case, the cylindrical ring model was applied: the galaxy was discretised into rings of different radii and the results compared with the observational data from Mollá et al. [1]. The radii chosen were $r \in [0.001, 22] \text{ kpc}$ in steps of 1 kpc.

The results for the stellar surface density (Σ_*) are presented in Figure 4, where it is clear that the model reproduces the observational data very well across the full radial range. This agreement confirms that the exponential infall profile, combined with the adopted disk parameters, correctly captures the integrated stellar mass build-up at each radius.

However, looking at the gas surface density in Figure 5, a gap between the model prediction and the observations is evident. This discrepancy is concentrated at small radii ($r \lesssim 5 \text{ kpc}$), which is expected since the model treats all rings as closed, independent systems with the same physical prescriptions and a spatially uniform accretion timescale τ . The central regions of the Milky Way host a variety of complex processes that are not captured by this approach: bar-driven radial gas inflows can efficiently funnel material towards the centre [25]; secular evolution of

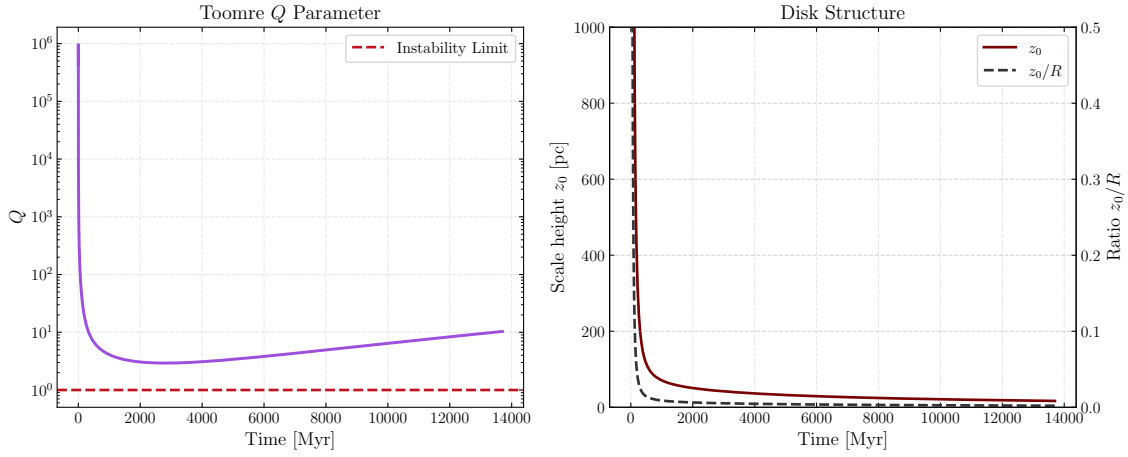


Figure 3: Diagnostic parameters for the constant infall model: Toomre stability parameter Q (left), disk scale height z_0 and relative disk thickness z_0/R (right). The red dashed line marks the instability limit $Q = 1$.

the bar and bulge modifies both the gas distribution and the star formation history [26]; and AGN feedback, while likely subdominant in the Milky Way, may contribute to suppressing star formation in the innermost regions [27]. All these processes tend to redistribute gas inward and suppress the central SFR beyond what a simple ring model predicts.

The star formation rate surface density (Σ_{SFR}) is shown in Figure 6. Consistently with the gas profile, the model agrees well at large radii but overestimates the observed values at small radii. This discrepancy suggests that the model’s limitations in the innermost regions stem from the assumption of a radially independent infall timescale (τ). In reality, a constant τ fails to capture the ‘inside-out’ formation scenario, where inner regions accumulate mass faster than the outskirts.

Also, the gas surface density and the star formation rate can be combined to compute the depletion time ($\tau_{\text{dep}} = \Sigma_g / \Sigma_{\text{SFR}}$). This quantity is an important timescale in astrophysics, indicating how long the galaxy would take to consume its gas reservoir at the current star formation rate. In spiral galaxies like the Milky Way, typical values are $\tau_{\text{dep}} \approx 1.5\text{--}2.0$ Gyr [28, 29], with some variation depending on the galactocentric radius and the local gas phase. The depletion time profile is shown in Figure 7, where the model agrees well with the observational data across most of the disk. The exception is, once again, the innermost radii, where the model underestimates τ_{dep} . This is consistent with the overestimation of the central gas and SFR densities discussed above: if the model predicts more gas and more star formation at small radii, the ratio $\Sigma_g / \Sigma_{\text{SFR}}$ is not necessarily well constrained there, and the systematic failure reinforces the conclusion that the complex dynamics of the bulge region are essential to explain the observed behaviour at $r \lesssim 5$ kpc.

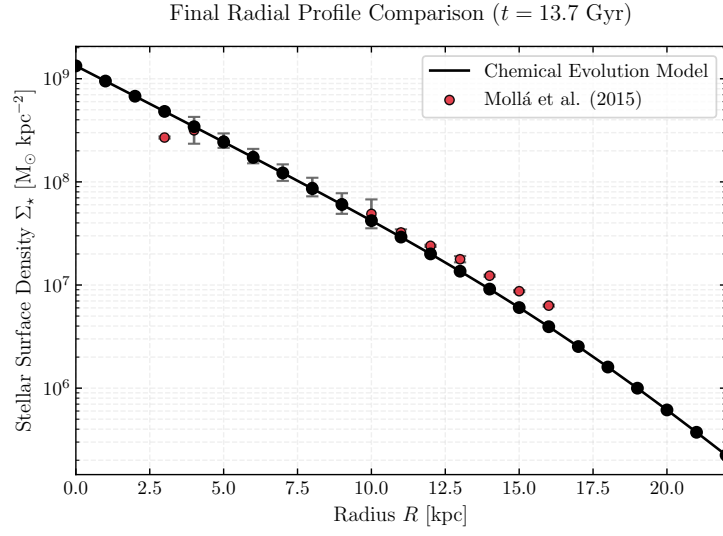


Figure 4: Stellar surface density Σ_* as a function of galactocentric radius. The solid line shows the model prediction and the points show the data from Mollá et al. [1].

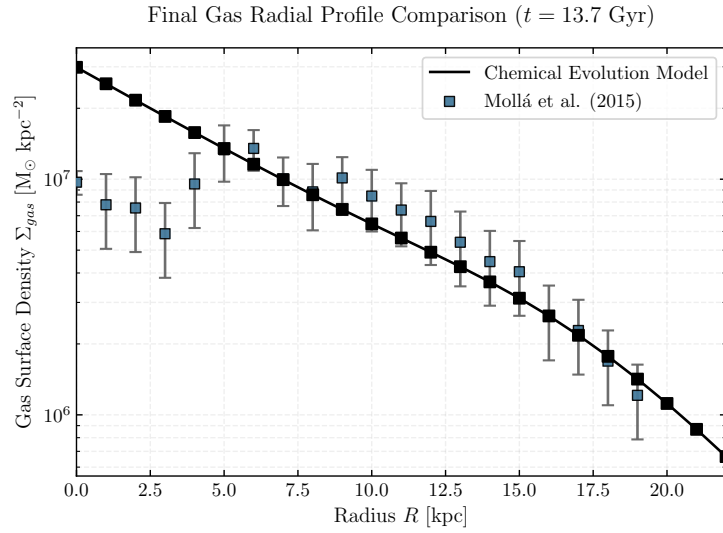


Figure 5: Gas surface density Σ_{gas} as a function of galactocentric radius. The solid line shows the model prediction and the points show the data from Mollá et al. [1].

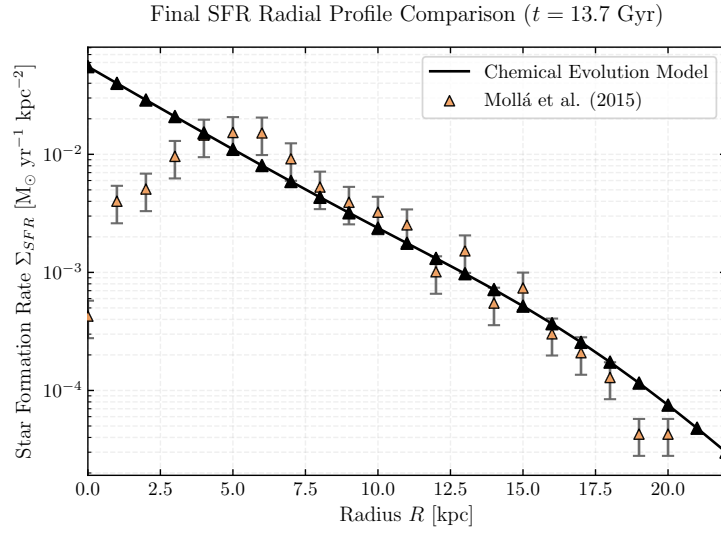


Figure 6: Star formation rate surface density Σ_{SFR} as a function of galactocentric radius. The solid line shows the model prediction and the points show the data from Mollá et al. [1].

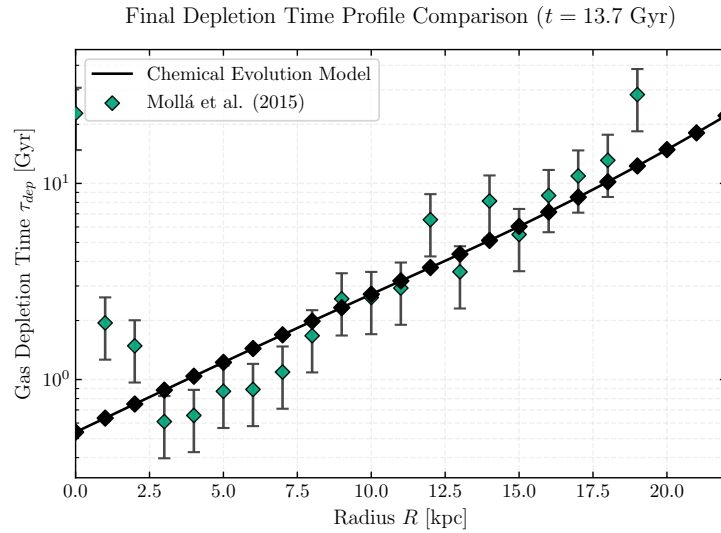


Figure 7: Gas depletion time $\tau_{\text{dep}} = \Sigma_g / \Sigma_{\text{SFR}}$ as a function of galactocentric radius. The solid line shows the model prediction and the points show the data from Mollá et al. [1].

3.4 Inside-out model with Milky Way parameters

Aiming for a more realistic case, and in an attempt to improve the agreement at small radii, the parameters were updated to values closer to those considered in Mollá et al. [1]. Specifically, $R_d = 3500$ pc, $\Sigma_0 = 1500 \text{ M}_\odot \text{ pc}^{-2}$, $\text{SFE} = 0.02$, and, initially, $\tau = 7000$ Myr were adopted. The inside-out formation scenario was then activated by letting τ vary with radius as $\tau = 1000 + 1500(r/\text{kpc})$ for $r \in [0.001, 22]$ kpc [12, 19].

The results are shown in Figure 8. The model reproduces the observed $\log \Sigma_*$ profile well across almost the full radial range. More importantly, the agreement between the predicted and observed $\log(\Sigma_{\text{SFR}})$ is notably better than in the previous model. This improvement can be understood as follows: allowing τ to increase with radius means that the inner rings accrete their gas on a shorter timescale (τ small) and therefore reach a higher peak SFR earlier, retaining less gas today. Conversely, the outer rings accrete gas more slowly, so they are still building up their stellar mass at the present time. This inside-out growth naturally produces a steeper radial gradient in Σ_* and a less centrally concentrated SFR profile, both of which are key features of observed Milky-Way-like galaxies [12, 1]. The remaining mismatch at $r \lesssim 1$ kpc persists because the radial gas flows and secular bar dynamics discussed in the previous section are still not included.

Furthermore, the mass evolution and velocity dispersion at $R = 8$ kpc are plotted in Figure 9 in order to study the mass structure and morphology of the galaxy. The gas mass reaches its maximum near 1000 Myr and then decreases, approaching $0 \text{ M}_\odot$, while the stellar mass grows continuously, although more slowly after 5000 Myr. This is an expected result from Equation (1.2): at early stages, ψ is small and $\dot{M}_g \approx I$, so the gas accumulates rapidly because the infall is exponential. Given that $\psi \propto M_g/t_{ff}$, the SFR increases as the gas builds up, until M_g reaches its maximum and the consumption rate exceeds the infall rate. This transition is clearly visible as a change of slope in the stellar mass curve. The velocity dispersion σ also reflects this behaviour: it rises almost instantaneously as the large initial gas infall compresses and stirs the interstellar medium, followed by a pronounced decline as the energy is progressively absorbed by stellar formation processes. Near 2000 Myr, σ reaches a local maximum, which can be interpreted as the moment when the energy injected by supernovae is highest, coinciding with the peak of the SFR.

In addition, the $\log \Sigma_*$ increases rapidly at early times and converges to a nearly constant value, indicating that the galaxy self-regulates its star formation as the gas is depleted. The disk scale height shows a fast decrease at early stages, consistent with the behaviour shown in Figure 10. From Equations (2.5) and (2.7), $z_0 = \sigma^2/g \propto \sigma^2/\Sigma$, so the rapid decrease of σ combined with the increase of g due to the growing mass drives a sharp fall of z_0 , which stabilises near $z_0/R \approx 0$. However, this result indicates that z_0 is lower than the values typically inferred from observations of

stellar disks, which range from ~ 100 to ~ 1000 pc [14]. Two simplifications adopted in this model contribute to this discrepancy. First, the plane-parallel potential assumes that stars and gas occupy the same thin layer, whereas in reality stars are distributed in a thicker disk than the gas, which provides additional vertical support [14]. Second, the Sedov-Taylor cooling prescription [17] assumes a power-law cooling function in the temperature range $T \in [10^5, 10^7]$ K with radiative losses $L \propto E$, without an upper ceiling or changes in behaviour at higher temperatures, which may underestimate the energy retained in the turbulent gas and hence underestimate σ at late times.

Allied to this, the Toomre parameter (Q) is shown in Figure 10. Q decreases during the first 1000 Myr as M_g grows rapidly: from Equation (1.8), $Q \propto \sigma/\Sigma_g$, and since Σ_g grows faster than σ at early times, Q drops towards its minimum, which coincides with the moment when M_g reaches its peak. After that, as the gas is consumed and Σ_g decreases, Q increases steadily, indicating that the disk becomes progressively more stable. Towards the end of the simulation, $Q \gg 1$, which means the galaxy is gravitationally stable but star formation will eventually cease as the gas reservoir is exhausted.

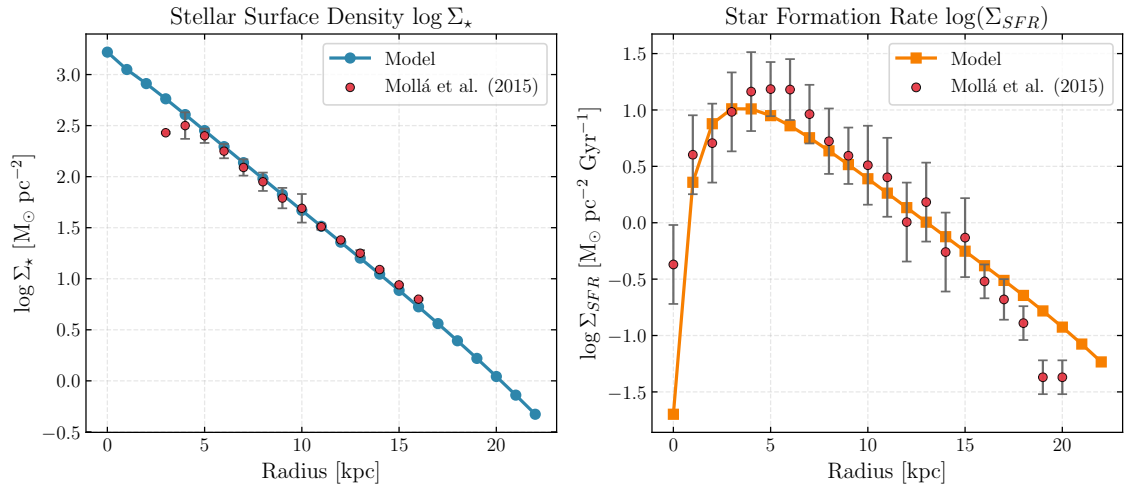


Figure 8: Radial profiles for the inside-out model: stellar surface density $\log \Sigma_*$ (left) star formation rate $\log(\Sigma_{\text{SFR}})$ (right). Points show the data from Mollá et al. [1]; solid lines show the model.

3.5 Extreme cases

To verify the behaviour of the model under extreme conditions, τ was set to two values: $\tau = 1000$ Myr (fast infall) and $\tau = 10^9$ Myr (quasi-constant infall), representing a scenario where the entire gas reservoir is accreted almost instantaneously versus one where it is delivered at a nearly steady rate over a Hubble time. The results are shown in Figures 11 and 12.

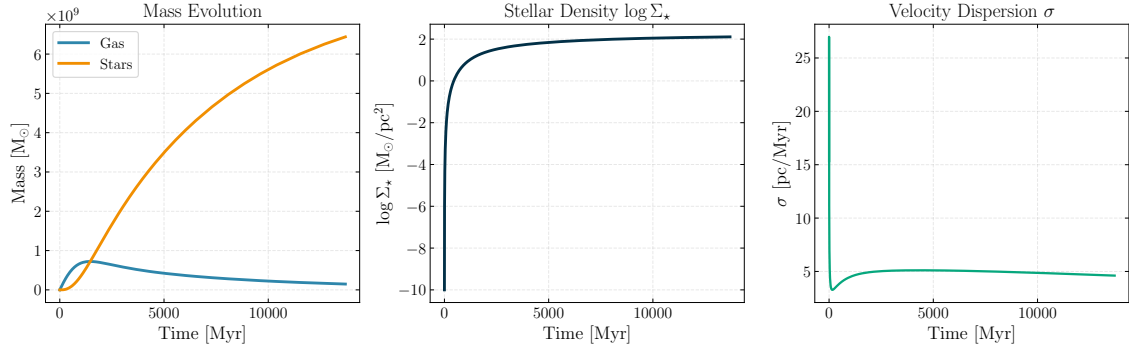


Figure 9: Time evolution at $R = 8$ kpc for the inside-out model: gas and stellar mass (left), stellar surface density $\log \Sigma_*$ (centre), and velocity dispersion σ (right).

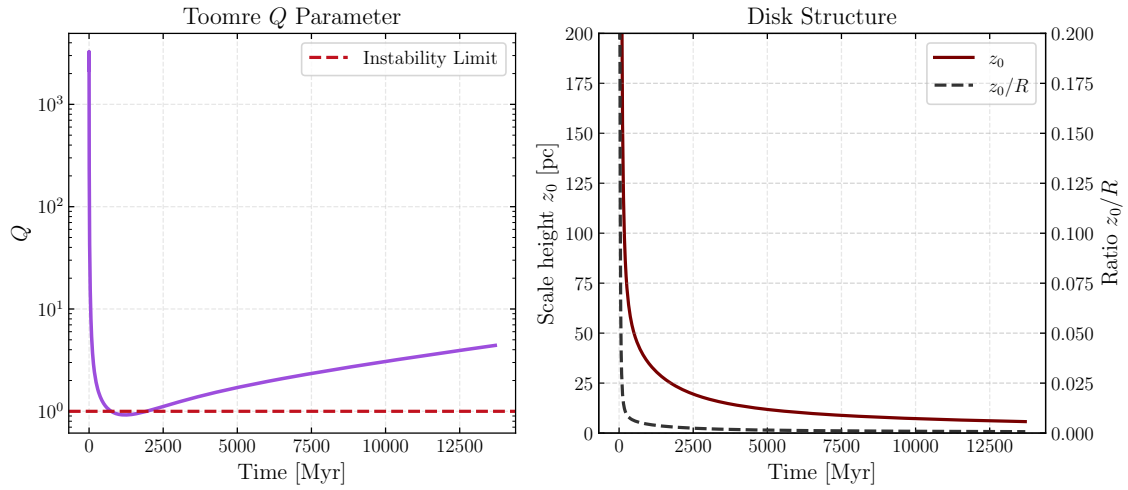


Figure 10: Dynamical diagnostics at $R = 8$ kpc for the inside-out model: Toomre stability parameter Q (left), with the red dashed line marking the instability limit $Q = 1$; disk scale height z_0 and relative thickness z_0/R (right).

As expected, a lower value of τ allows the gas and stellar mass to reach higher values in less time. Since the infall rate $I = I_0 e^{-t/\tau}$ peaks sharply at $t = 0$ for small τ , the gas reservoir fills rapidly and so does the SFR, given that $\psi \propto M_g/t_{ff}$. This leads naturally to an early maximum of star formation and, consequently, to a thinner disk at early times: from $z_0 = \sigma^2/g$ and $g \propto \Sigma \propto M_{\text{total}}/A$, the rapid mass accumulation increases g faster than σ can respond, driving z_0 downward. As shown in Figure 11, the quasi-constant infall case shows a much smoother evolution in mass quantities, since the gas is replenished at a nearly steady rate, preventing both a strong early burst and a sharp subsequent decline.

Furthermore, the dynamical comparison in Figure 12 reveals a qualitative difference in the stability history of the two scenarios. In both cases, Q decreases rapidly during the first ~ 500 Myr as Σ_g grows. For the fast-infall scenario, Q drops below unity, signaling a brief episode of gravitational instability that coincides with the peak of the Star Formation Rate (SFR). Subsequently, as gas is rapidly consumed and converted into stars, Σ_g decreases and Q recovers sharply, reaching values as high as 10^2 by the end of the simulation. This indicates a highly stable, gas-poor disk in the late stages of evolution.

In contrast, for the constant-infall model, Q remains near unity ($Q \approx 1$) throughout most of the simulation. This behavior indicates a disk that is marginally stable at all times, which is qualitatively consistent with the self-regulated star formation framework [8, 7], where the balance between gas accretion and feedback maintains the system near the stability threshold. This contrast suggests that the infall timescale plays a decisive role in setting not only the star formation history but also the structural and dynamical properties of the resulting disk. The fast-infall scenario is more representative of the rapid assembly of massive galaxies at high redshift [12, 19], where early, intense accretion episodes drive starbursts and lead to a rapid settling of the galactic disk.

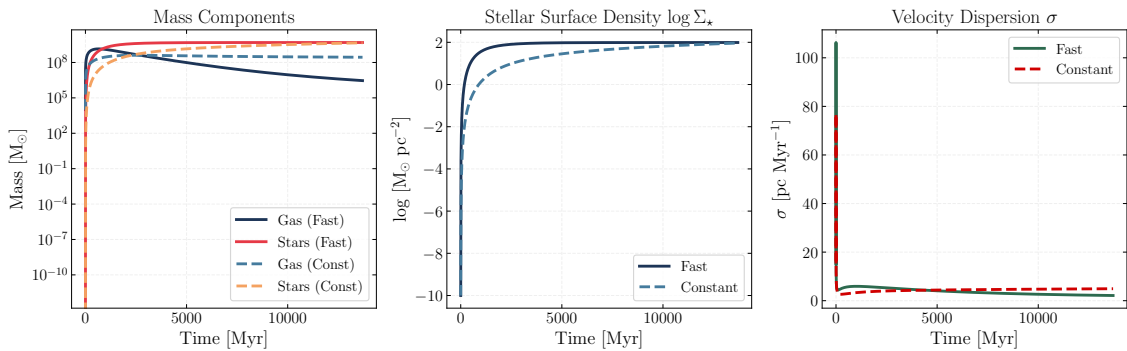


Figure 11: Comparison of the fast ($\tau = 1000$ Myr) and quasi-constant ($\tau = 10^9$ Myr) infall cases: gas and stellar mass (left), stellar surface density $\log \Sigma_*$ (centre), and velocity dispersion σ (right), all as a function of time.

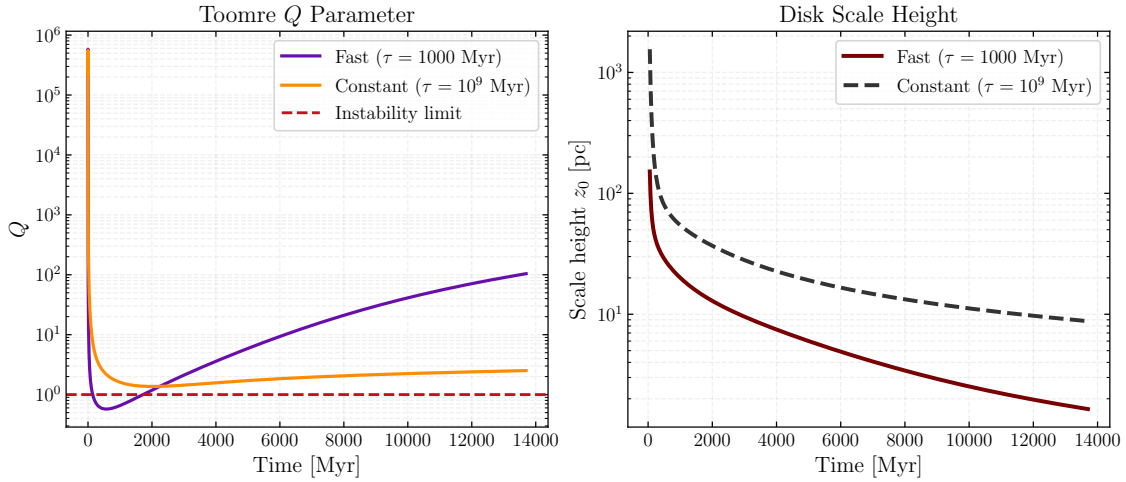


Figure 12: Comparison of the fast ($\tau = 1000$ Myr) and quasi-constant ($\tau = 10^9$ Myr) infall cases: Toomre stability parameter Q (left), with the red dashed line marking the instability limit $Q = 1$; disk scale height z_0 and relative thickness z_0/R (right).

4 Conclusions

4.1 Main results

The model approximation of cylindrical shells with the gravitational field of an infinite plane (2.1), hydrostatic equilibrium (2.4), energy equipartition, a constant SFE, and the implementation of energy dissipation through a radiative cooling timescale t_c derived from the Sedov-Taylor analytical framework (2.16), proved to be highly consistent, as evidenced by the close agreement between the simulated radial profiles and the observational data from Mollá et al. [1] across most of the galactic disk. The stellar surface density, gas surface density, star formation rate, and gas depletion time are all reproduced well at radii $r \gtrsim 5$ kpc, confirming that the key physical ingredients—exponential gas infall, gravitational support by both gas and stars, and supernova feedback—are sufficient to describe the outer and intermediate disk.

At small radii ($r \lesssim 5$ kpc), the initial exponential infall model with a fixed τ systematically overestimates the observed gas and stellar surface densities. This occurs because the exponential surface density profile $\Sigma_0 e^{-r/R_d}$ deposits more mass per unit area at small radii, and the closed-ring prescription has no mechanism to redistribute this gas outwards, causing it to accumulate and be converted into stars at an excess rate. This discrepancy motivated the implementation of the inside-out scenario through a radially dependent timescale $\tau(r)$ [12, 19]: since the total accreted mass per ring scales as $M_{\text{total}} \propto I_0 \cdot \tau$, a shorter τ at small radii reduces the central mass budget, bringing the predicted profiles significantly closer to the observations across the intermediate disk.

4.2 Limitations of the model

However, at the innermost radii ($r \lesssim 1$ kpc), the inside-out model still overestimates the stellar surface density while simultaneously underestimating the current star formation rate. These two discrepancies are physically consistent with one another: the model depletes its gas reservoir at early times, forming many stars but leaving too little gas to sustain the observed SFR today. In the real galactic centre, this depletion does not occur to the same extent because bar-driven radial inflows continuously replenish the central gas supply [25], maintaining an elevated SFR even after a long history of star formation. This replenishment mechanism is intrinsically non-local and beyond the scope of the closed-ring approximation. Additional processes that further regulate the central gas cycle include the secular evolution of the bar and pseudobulge [26] and AGN feedback, which can suppress star formation in the most gas-rich phases by driving outflows. Incorporating radial gas flows and a separate bulge component would be the most direct path to improving the model in this regime.

Two further structural limitations of the current framework deserve attention. First, the model yields disk scale heights z_0 well below the observed range of 100–1000 pc [14]. This points to the failure of the assumption that stars and gas occupy the same infinitesimally thin plane: in reality, the stellar disk is significantly thicker than the gas layer, and its gravitational contribution modifies the vertical equilibrium, leading to larger scale heights z_0 [14]. Implementing a two-component geometry—a thin gas disk embedded in a thicker stellar disk—would be a natural extension.

Second, the cooling timescale t_c adopted in this work is derived from the analytical Sedov-Taylor solution [17] and constitutes an approximation in several respects. The Sedov-Taylor phase strictly describes the adiabatic stage of a supernova remnant expanding into a uniform ambient medium, during which radiative losses are negligible; t_c is then defined as the moment when the cooling time of the post-shock gas equals the age of the remnant, marking the transition to the radiative phase. This transition timescale is adopted in our model as a global energy dissipation rate $L = E/t_c$, which is a simplification: it replaces the actual time-dependent cooling process with a single characteristic timescale. Furthermore, the cooling function employed, $\Lambda(T) \approx \Lambda_a (T/T_a)^a$, is calibrated for a gas of solar metallicity with a fixed ambient density $n_0 = 1 \text{ cm}^{-3}$ [17]. In reality, the ISM metallicity is not constant. A more realistic treatment would require a metallicity-dependent cooling function, such as those tabulated in McKee and Ostriker [13] or modern multi-phase ISM models [8], which would allow t_c to vary both with radius and with the evolving chemical enrichment of the gas.

4.3 Future Improvements

Furthermore, a more realistic model should treat the star formation efficiency not as a fixed parameter but as a quantity determined by the fraction of gas that exceeds the local collapse threshold. A more physically motivated criterion is the Bonnor–Ebert mass [10], which describes the maximum mass of a pressure-confined, isothermal gas sphere in hydrostatic equilibrium:

$$M_{\text{BE}}(\rho) = 1.182 \frac{\left(c_s^2 + \frac{3}{2}v_A^2\right)^{3/2}}{\sqrt{G^3\rho}}$$

where c_s is the isothermal sound speed and v_A the Alfvén velocity. The Bonnor–Ebert mass explicitly accounts for the confining pressure of the surrounding medium and for magnetic support through v_A , making it more appropriate for dense cloud cores embedded in a turbulent ISM. Moreover, since the gas density distribution follows a log-normal function whose width is set by the Mach number $\mathcal{M} = \sigma/c_s$ [9], the collapsing mass fraction—and hence the SFE—depends directly on σ . Adding this dependence would naturally couple the SFE to the local turbulent state of the gas, allowing it to vary self-consistently with radius and potentially resolving the residual mismatch in the predicted SFR profile at the innermost radii. In practice, this would require promoting the collapsed mass fraction to a dynamical variable in the ODE system, making the model fully self-regulating.

Declaration on the use of AI tools

Artificial intelligence tools were used during the preparation of this work exclusively for language refinement, academic writing assistance, bibliography search support, and code debugging. All physical interpretations, numerical implementations, analysis of results, and scientific conclusions were developed and verified by the author.

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